

Introduction & Hook

Ask:

“How are high-end VR simulations like flight simulators or military training systems created?”

Explain:

Such systems are built using **advanced simulation tools**, not just game engines.

What are Vega and MultiGen Tools?

Overview

MultiGen (Creator)

A 3D modeling tool used for creating real-time simulation environments.

- Used for:
 - o Terrain modeling
 - o Buildings
 - o Military environments
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Vega

A real-time simulation engine used to run and visualize virtual environments.

- Works with MultiGen models
 - Handles:
 - o Rendering
 - o Interaction
 - o Simulation
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Key Relationship:

MultiGen → Creates models
Vega → Runs simulation

MultiGen (Creator) – Scene Creation

Scene Creation Process

Steps in Scene Creation:

1. Create terrain
 2. Add objects (buildings, vehicles)
 3. Apply textures
 4. Define hierarchy
 5. Export scene
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Features:

- Real-time modeling
 - LOD support
 - Scene graph structure
 - Database creation
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Output Format:

- OpenFlight (.flt) format
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Example:

Creating a virtual airport:

- Runway
- Control tower

- Aircraft models
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Vega – Simulation Engine

Vega Features

Functions:

- Real-time rendering
 - Physics simulation
 - Interaction handling
 - Camera control
 - Lighting
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Capabilities:

- High-performance rendering
 - Supports large-scale environments
 - Multi-channel display support
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Example:

Flight simulator:

- Pilot interacts with controls
 - System updates environment in real time
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Integration of MultiGen and Vega



Workflow

Model Creation (MultiGen)



Export (OpenFlight)



Load into Vega



Run Simulation

Key Idea:

- MultiGen = Design
 - Vega = Execution
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Simulation Applications

Real-World Applications

1.Military Training

- Battlefield simulation
 - Tank/aircraft training
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2.Flight Simulation

- Pilot training
 - Aircraft testing
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3. Urban Planning

- Smart city simulation
 - Traffic modeling
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4. Industrial Training

- Machine operation
 - Safety training
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Key Insight:

These tools are used where **accuracy and realism are critical**.

Comparison with Game Engines

Feature	Vega/MultiGen	Unity/Unreal
Purpose	Simulation	Games/VR apps
Accuracy	High	Moderate
Cost	Expensive	Affordable
Use case	Defense/industry	General VR

Recap & Questions

Key Points:

- MultiGen is used for modeling
 - Vega is used for simulation
 - Together they create real-time VR systems
 - Used in high-end professional applications
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Quick Questions:

1. What is MultiGen used for?
2. What is Vega?
3. What is OpenFlight format?

4. Give examples of simulation applications